

BXP Portfolio Summary

Field | Detail

Project | Breathe Exposure Protocol (BXP)
Research area | Open data standards; atmospheric exp
Duration | February 2026 – present (~6 months a
Capacity | Independent research and development
Repository | <https://github.com/bxpprotocol/bxp-s>
DOI (specification) | <https://doi.org/10.5281/zenodo.18906>
DOI (implementation) | <https://doi.org/10.5281/zenodo.18907>
License | Apache 2.0

Problem

Air pollution causes approximately 7 million premature deaths annually. Affordable sensors now exist across the world, but the data they produce is fragmented: every manufacturer, agency, and research network uses incompatible formats. Data that cannot communicate with itself cannot coordinate a response. The barrier is not hardware — it is the absence of a common data standard.

Research

Identified the air quality data interoperability problem through independent investigation. Surveyed existing standards (US AQI, European CAQI, OpenAQ, OGC SensorThings, HL7 FHIR) and found none adequate for universal, open, privacy-preserving atmospheric exposure data exchange. Studied WHO Air Quality Guidelines 2021 and DALY burden data for the health risk index design. Investigated cryptographic techniques for data integrity. Reviewed GDPR, CCPA, and POPIA privacy frameworks.

Technical Work

Artefact | Scale | Status
BXP v2.0 Protocol Specification | 1,300 lines | Complete
Reference Server (Python/FastAPI) | 1,877 lines | Complete
Python SDK | 895 lines | Complete
TypeScript SDK | ~400 lines | Complete
CLI Tool | 740+ lines | Complete
MQTT Bridge | ~200 lines | Complete
Sample dataset (10 global cities) | 250 lines JSON | Complete
Documentation (API, guide, architect | ~1,000 lines | Complete

My Contribution

Sole creator of every component listed above. Conceived the protocol concept, identified the interoperability problem, designed the architecture, wrote the specification, built the

reference implementation, and published both to Zenodo. The project began as an investigation and evolved through iteration: v1.0 (February 2026) → systematic 50-item gap analysis → v2.0 specification → v2.1 reference server addressing all identified gaps.

Key Skills Demonstrated

- * Protocol design and formal specification writing
 - * Open data standards development
 - * Systems architecture (data pipeline, privacy framework, federated network design)
 - * Full-stack software development (Python backend, REST API, web dashboard)
 - * Cryptographic integrity and privacy engineering (SHA-256, k-anonymisation, geohash)
 - * Developer tooling (SDK design, CLI, IoT integration)
 - * Scientific methodology (structured problem identification, gap analysis, honest limitation acknowledgement)
 - * Technical communication (specification, documentation, evidence packaging)
-

Current Status

BXP v2.0 is a complete, publicly available open protocol specification with a working reference implementation, two SDKs, a CLI tool, ecosystem tooling, and permanent DOIs. The protocol addresses a genuine, documented gap in air quality data infrastructure. The reference implementation is a functional prototype — not a production system, not externally adopted, but fully buildable and demonstrably functional.

Evidence Available

Evidence | Strongest for
SPEC.md – formal specification with | Protocol design, systems thinking
Reference server source code | Implementation ability
Python SDK | Software engineering
CHANGELOG.md with v1.0 → v2.1 histor | Iteration and growth
Development gap analysis (50 items) | Systematic, self-critical thinking
Sample dataset (6 continents) | Global scope
Zenodo DOIs | Independent publication, permanent r
GitHub repository | Open-source, reproducible, public

Honest Limitations

The binary file format is specified but not yet implemented. The federated network is designed but not yet built. No third-party has independently implemented BXP. BXP_HRI has not been clinically validated. The BXP Foundation does not yet exist. These are the natural boundaries of independent research by one person.

The air is public. The data should be too.

Copyright 2026 Elvarin — Apache 2.0